# Fast guide for ANPR of TCG camera via ISAPI

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# Description

This manual defines the HTTP-based OPEN Intelligent Security Application Programming Interface (ISAPI).The OPEN Intelligent Security API is used for the intelligent security clientsoftware or system to communicate with the devices.

# Authorization and WWW-Authentication

The request sent to device should be authenticated by the Digest Access method of the RFC 2617 protocol, so all the devices should support the Digest Access method.
Each request must be sent with Authorization header field. Otherwise, HTTP response (401) will be returned with WWW-Authenticate header field.
For details about the Digest Authentication, refer to the website:
http://www.ietf.org/rfc/rfc2617.txt.

# How to get real time LPR data via ISAPI

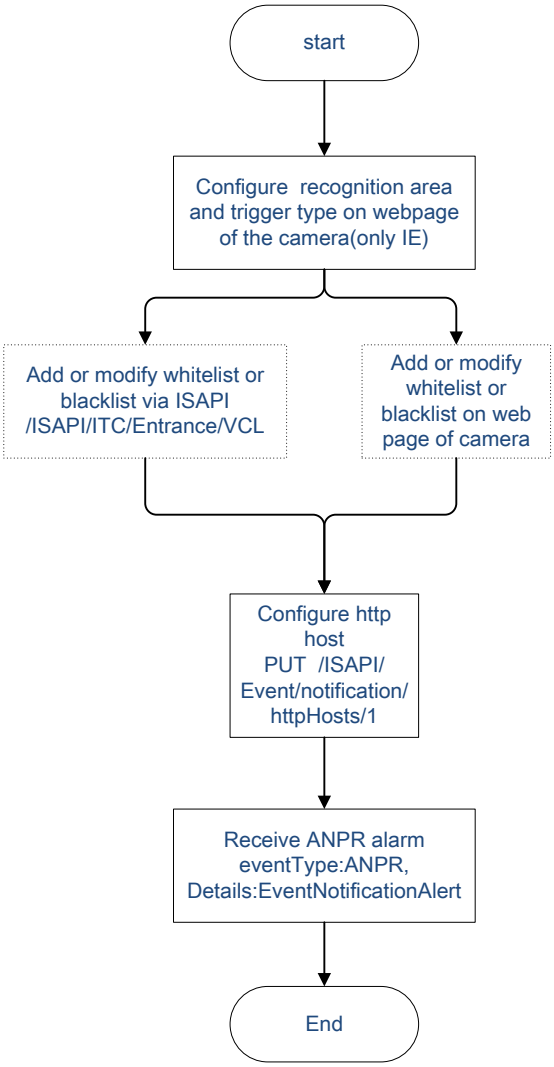
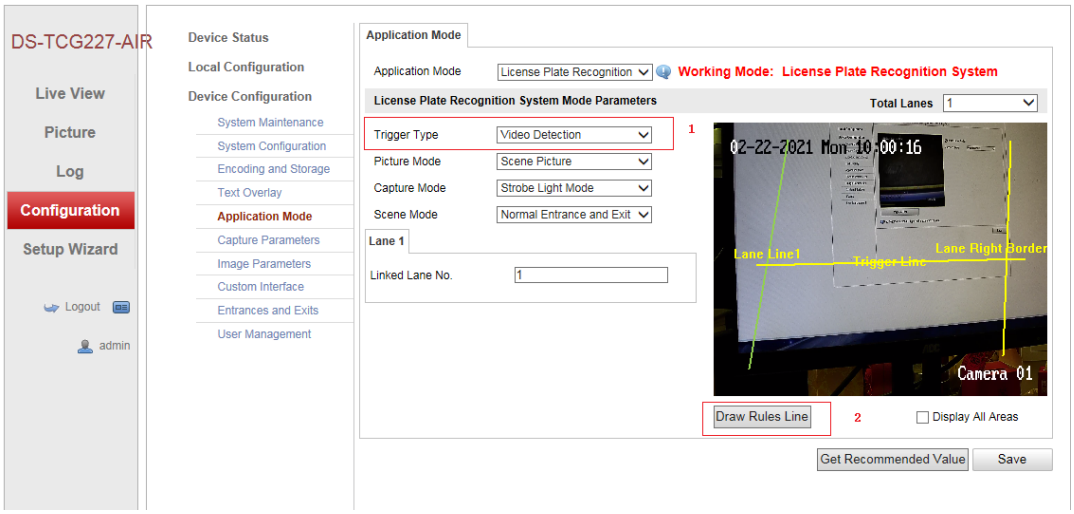



Figure 1 Programming flow of configuring ANPR alarm

For detailed ISAPI protocol or operation on camera, please refer to )1, 2), 3) and 4)

## 1)configure trigger type and recognition area

Please install plugin on IE exploer while login the camera. In order to recognize license plate, detection area must configured at first. Please configure **trigger type**[1] and **recognition area**[2] on the web page of the camera.

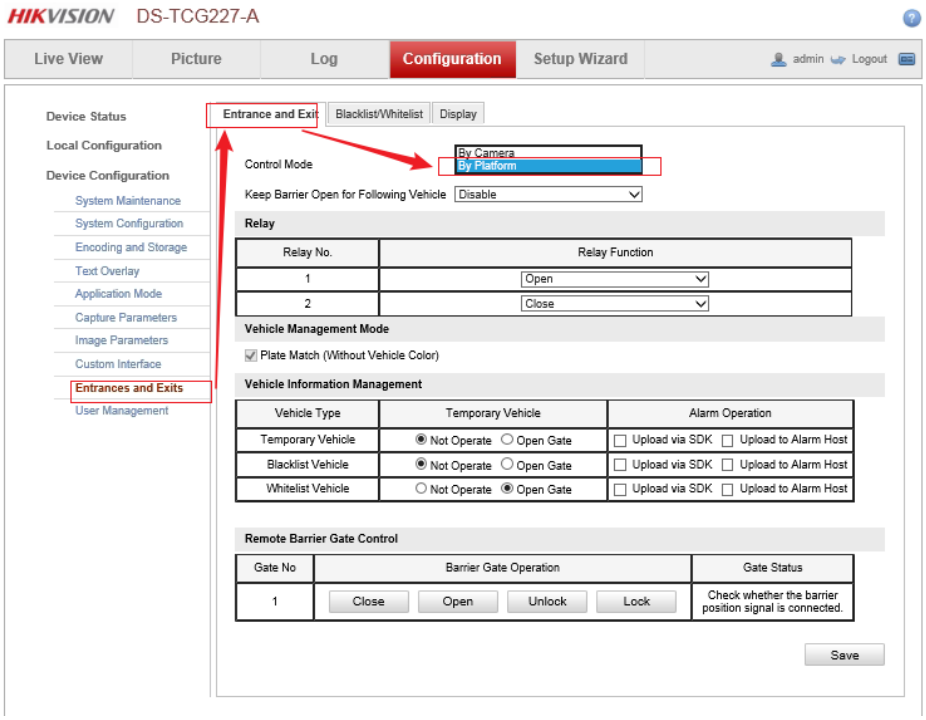



# 2)configure white/black list

We have two ways to control barrier. The first one is control barrier by your platform and the sencond one is control barrier by the camera. If your platform needs to control barrier by the camera(white/black list), please go to “Control barrier by the license plate stored in the camera” directly.

## Control barrier by platform

If your platform needs to control the barrier by itself, please follow below configuration on web page of the camera.

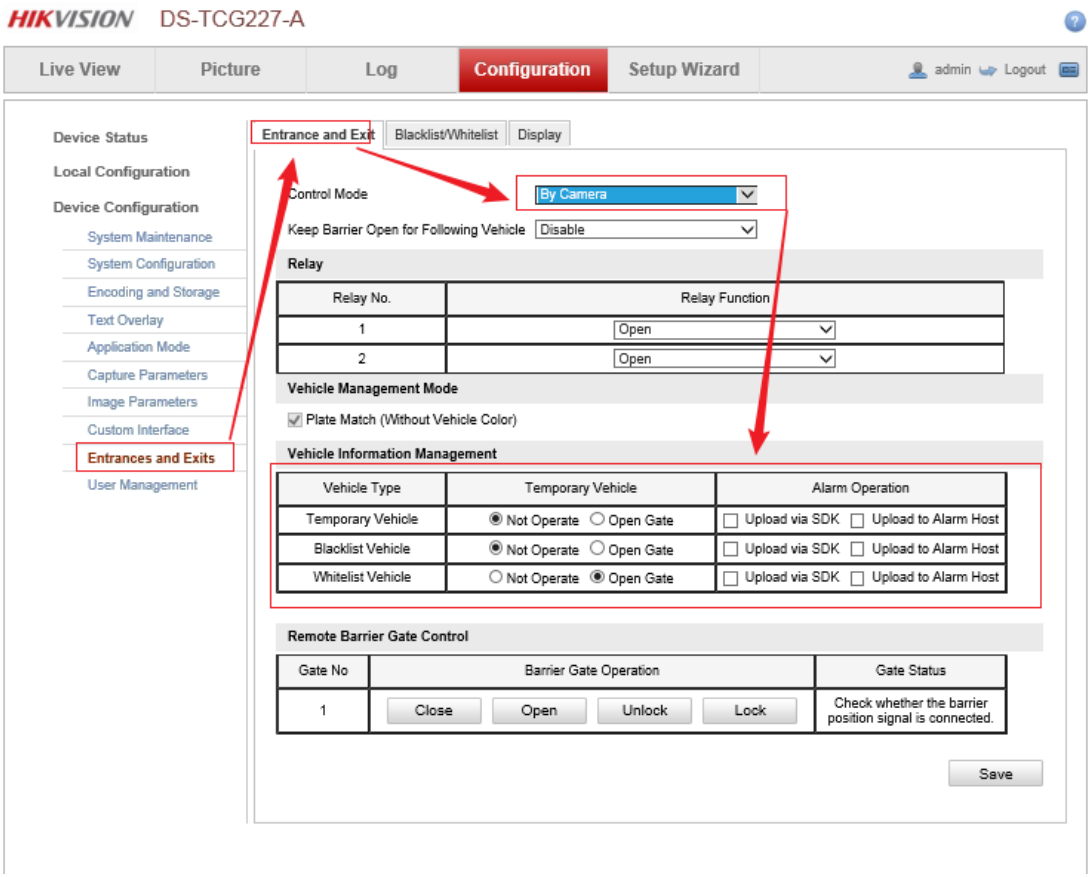



please use below ISAPI command to control the barrier

| | |
|---|---|
| **URL:** | /ISAPI/ITC/Entrance/barrierGateCtrl |
| **Method:** | PUT |
| **Description:** | It is used to control gate switch and lock |
| **Authentication** | Digest |
| **Query Parameters:** | None |
| **Inbound Data:** | <?xml version="1.0" encoding="utf-8"?><br><BarrierGateCtrl version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema"><br><barrietGateNum><!--opt,xs:integer,"1~2",barrier number--></barrietGateNum><br><BarrierGateCtrlList><br><br><barrietGateOper><!--opt,xs:string,"off,on,stop,locked",off-close the barrier,on-open the barrier,stop- stop barrier while in the process of close or open,locked-lock the barrier --></barrietGateOper><br></BarrierGateCtrlList><br></BarrierGateCtrl> |
| **Return:** | |

# Control barrier by the license plate stored in the camera

If your platform needs to control barrier by the license plate stored in the camera,please read the following part. Please follow below configuration on webpage of the camera.




You can configure white/black list via ISAPI protocol or on web page of the camera.

ISAPI protocol:

a. add license plate one by one to the camera

| | |
|---|---|
| **URL:** | /ISAPI/ITC/Entrance/VCL |
| **Method:** | PUT |
| **Description:** | Add or modify one license plate to list |
| **Authentication** | Digest |
| **Query Parameters:** | None |
| **Inbound Data:** | <?xml version="1.0" encoding="utf-8"?><br><SetVCLData><br><VCLDataList><br><singleVCLData><br><id>0</id><br><runNum>0</runNum><br><listType>0</listType> |

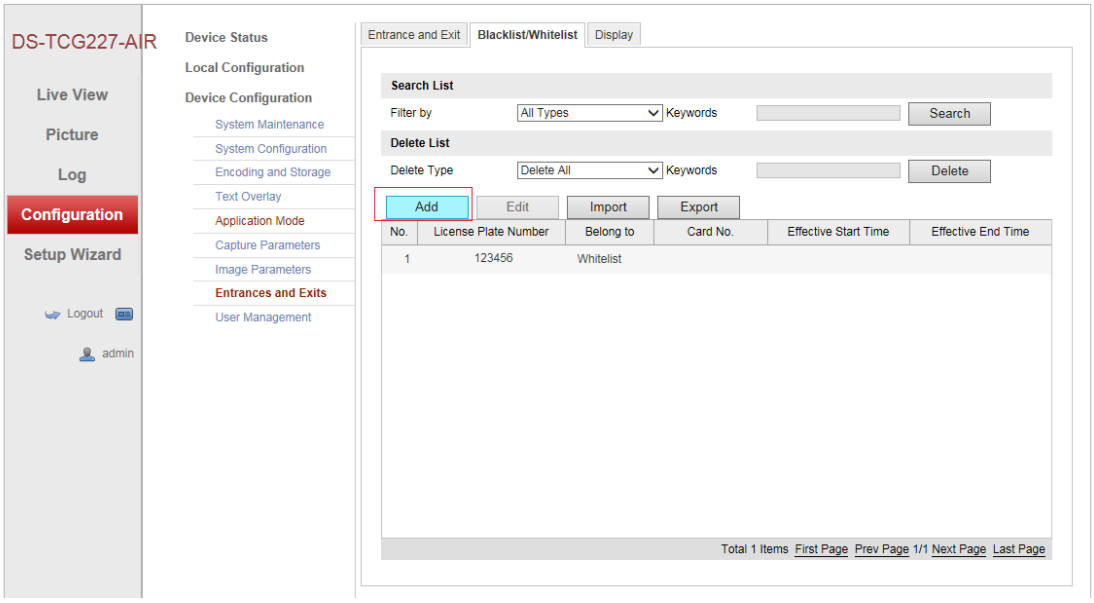
| | |
|---|---|
| | <plateNum>123</plateNum><br><cardNo/><br><startTime>0000-00-00T00:00:00Z</startTime><br><endTime>0000-00-00T00:00:00Z</endTime><br></singleVCLData><br></VCLDataList><br></SetVCLData><br><br>Note: value of listType<br>0:Whitelist<br>1:Blacklist<br>2:Graylist<br>3:Yellolist<br>4:Otherlist |
| **Return:** | |

b. delete license plate one by one to the camera

| | |
|---|---|
| **URL:** | /ISAPI/ITC/Entrance/VCL |
| **Method:** | DELETE |
| **Description:** | Delete one license plate to list |
| **Authentication** | Digest |
| **Query Parameters:** | None |
| **Inbound Data:** | <?xml version="1.0" encoding="utf-8"?><br><VCLDelCond><br><delVCLCond>1</delVCLCond><br><plateNum>123</plateNum><br><plateColor>0</plateColor><br><plateType>0</plateType><br><cardNo>123</cardNo><br></VCLDelCond><br><br>Note: delVCLCond (value is 1) and plateNum are necessary. |
| **Return:** | |

Add license plate on webpage of the camera:

Our camera also support above function on webpage of the camera. The webpage also use the same ISAPI protocol. You can use F12 on browser to check the related ISAPI protocol



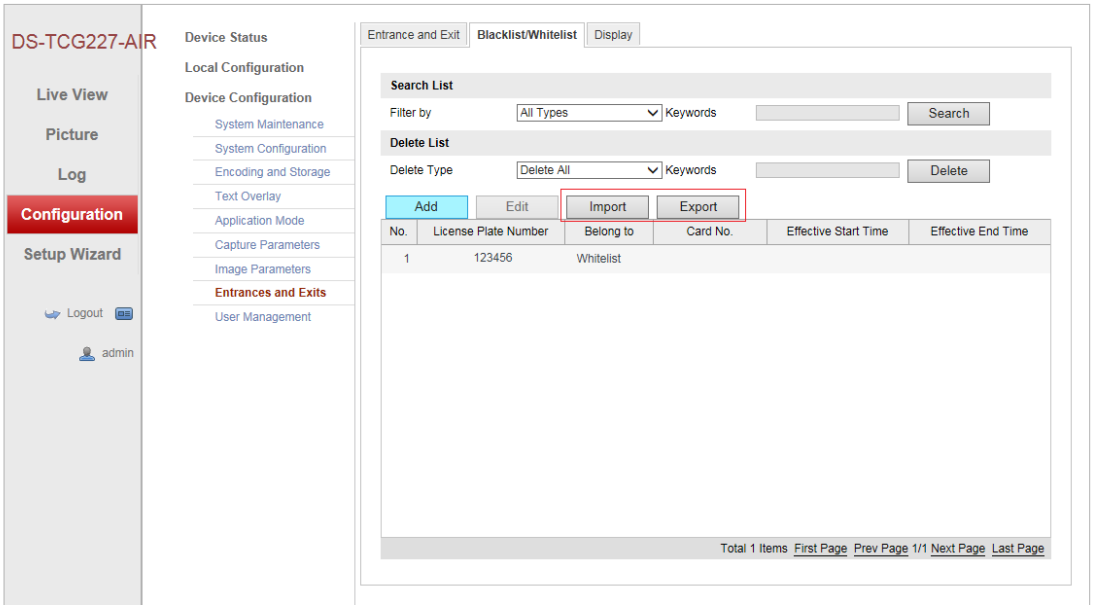

c. add license plate in batch :

add license plate in batach by ISAPI protocol:

Step1:

Export list (excel file) on webpage of camera at first. There is **no** ISAPI protocol for this function.

Sorry that our camera only support export license plate by IE plugin.




Step 2

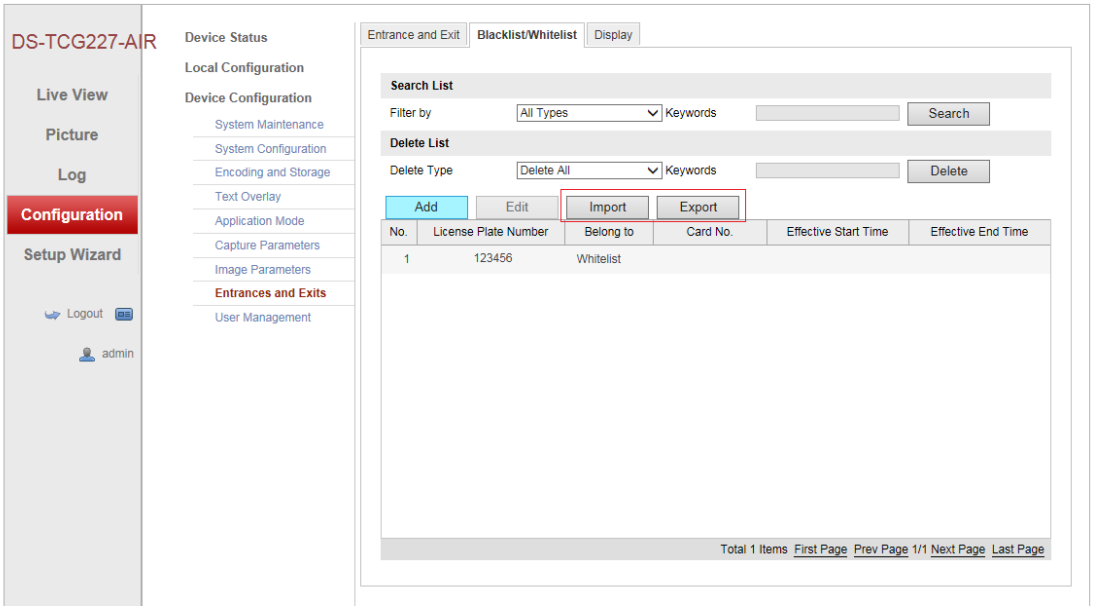
Modify the license plate list and send the file to the camera by below ISAPI command.

| | |
|---|---|
| **URL:** | /ISAPI/ITC/Entrance/VCL |
| **Method:** | POST |
| **Description:** | Import license plate list |
| **Authentication** | digest |
| **Query Parameters:** | None |
| **Inbound Data:** | Excel file exported in step 1 in binary |
| **Return:** | |

add license plate in batach on webpage of the camera:

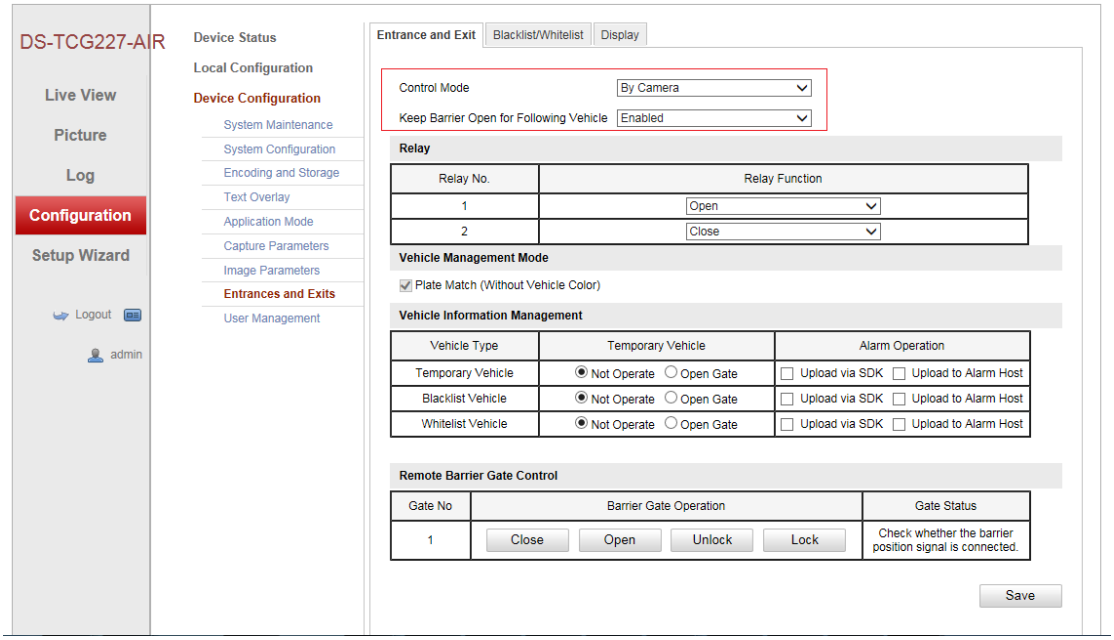
Our camera also support above function on webpage of the camera. You can import and export license plate on web page of camera:

Please refer to the webpage of the camera for configuratioin




If you have already configured white/black list to the camera, you can also configure the camera to control the gate according to the white/black list on webpage of the camera.

Select the control mode as “By Camera” and choose “Not Operate” or “Open Gate” for the temperory vehicle, balcklist vehicle and whitelist vehicle.




# 3)set HTTP host information

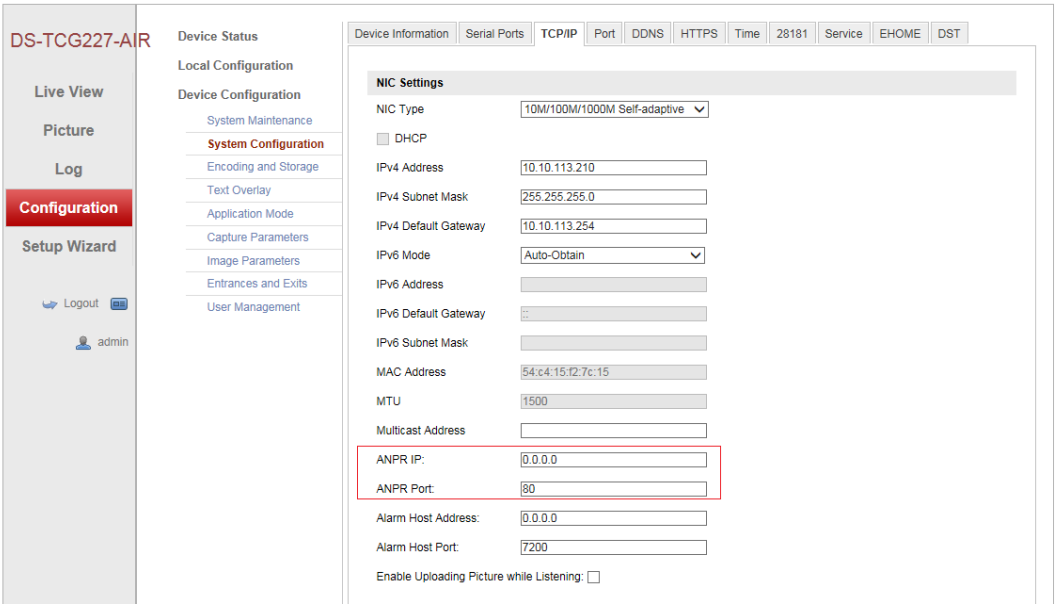
your platform is the HTTP server, our camera will send ANPR information via HTTP request with POST method to your platform.

| | |
|---|---|
| **URL:** | /ISAPI/Event/notification/httpHosts/1 |
| **Method:** | PUT |
| **Description:** | set HTTP host information |
| **Authentication** | digest |
| **Query Parameters:** | None |
| **Inbound Data:** | <?xml version="1.0" encoding="UTF-8"?><br><HttpHostNotification version="2.0"<br>xmlns="http://www.isapi.org/ver20/XMLSchema"><br><id>1</id><br><url>/test</url><br><addressingFormatType>ipaddress</addressingFormatType><br><ipAddress>1.1.1.3</ipAddress><br><portNo>8031</portNo><br></HttpHostNotification> |
| **Return:** | |

Note:

please modify ipAddress, portNo and url to your platform. Our cmaera will send ANPR data to your platform by HTTP URL: http://ipAddress:portNo/url with HTTP POST method.

If the http host is configured successfully, you can check the configuration on the web page of the camera




# 4)Parse ANPR alarm

when ANPR alarm is triggered, your platform should receive ANPR alarm(includes license plate picture, background picture and ANPR information), like below(only shows ANPR information):

POST /test HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: zh-CN
Content-Type: multipart/form-data;boundary=--------------------------7e13971310878
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 1.1.1.1:7200
Content-Length: 128512
Connection: Keep-Alive
Cache-Control: no-cache

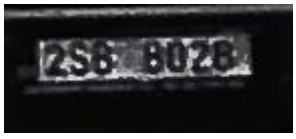
----------------------------7e13971310878
Content-Disposition: form-data; name="anpr.xml";filename="anpr.xml";
Content-Type: text/xml
Content-Length: 2269

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
<ipAddress>10.10.114.90</ipAddress>
<ipv6Address>::</ipv6Address>

<portNo>12674</portNo>
<protocol>HTTP</protocol>
<macAddress>54:c4:15:f2:a7:25</macAddress>
<channelID>1</channelID>
<dateTime>2019-03-01T14:21:35+08:00</dateTime>
<activePostCount>1</activePostCount>
<eventType>ANPR</eventType>
<eventState>active</eventState>
<eventDescription>ANPR</eventDescription>
<channelName>IP CAPTURE CAMERA</channelName>
<ANPR>
<licensePlate>2SB8028</licensePlate>
<line>1</line>
<confidenceLevel>97</confidenceLevel>
<plateType>unknown</plateType>
<plateColor>other</plateColor>
<licenseBright>143</licenseBright>
<pilotsafebelt>unknown</pilotsafebelt>
<vicepilotsafebelt>unknown</vicepilotsafebelt>
<pilotsunvisor>unknown</pilotsunvisor>
<vicepilotsunvisor>unknown</vicepilotsunvisor>
<envprosign>unknown</envprosign>
<dangmark>unknown</dangmark>
<uphone>unknown</uphone>
<pendant>unknown</pendant>
<tissueBox>unknown</tissueBox>
<label>unknown</label>
<decoration>unknown</decoration>
<plateCharBelieve>99,99,98,92,99,93,99</plateCharBelieve>
<speedLimit>0</speedLimit>
<illegalInfo>
<illegalCode>0</illegalCode>
</illegalInfo>
<vehicleType>vehicle</vehicleType>
<featurePicFileName>1</featurePicFileName>
<detectDir>0</detectDir>
<detectType>0</detectType>
<barrierGateCtrlType>0</barrierGateCtrlType>
<alarmDataType>0</alarmDataType>
<dwIllegalTime>0</dwIllegalTime>
<vehicleInfo>
<index>0</index>
<vehicleType>1</vehicleType>
<colorDepth>0</colorDepth>

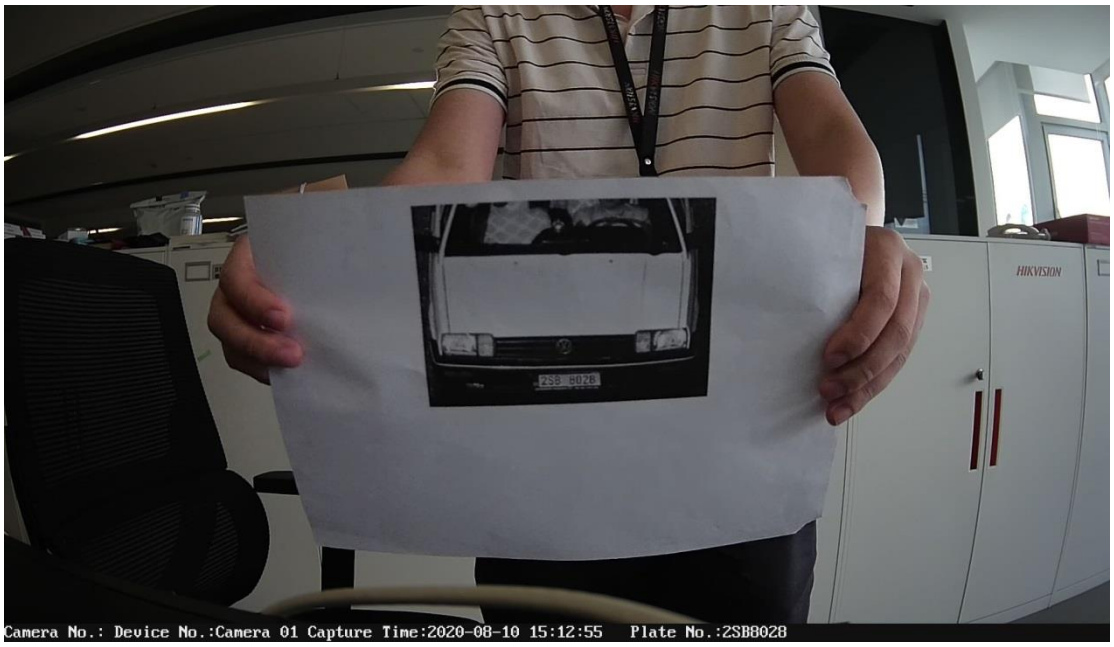
<color>unknown</color>
<speed>0</speed>
<length>0</length>
<vehicleLogoRecog>0</vehicleLogoRecog>
<vehileSubLogoRecog>0</vehileSubLogoRecog>
<vehileModel>0</vehileModel>
</vehicleInfo>
<pictureInfoList>
<pictureInfo>
<fileName>licensePlatePicture.jpg</fileName>
<type>licensePlatePicture</type>
<dataType>0</dataType>
</pictureInfo>
<pictureInfo>
<fileName>detectionPicture.jpg</fileName>
<type>detectionPicture</type>
<dataType>0</dataType>
<plateRect>
<X>496</X>
<Y>808</Y>
<width>66</width>
<height>24</height>
</plateRect>
</pictureInfo>
</pictureInfoList>
</ANPR>
<UUID>53845e3e-1dd2-11b2-9fe9-932e28ec1803</UUID>
<picNum>2</picNum>
</EventNotificationAlert>

----------7e13971310878
Content-Disposition:
form-data;name="licensePlatePicture.jpg";filename="licensePlatePicture.jpg";
Content-Type: image/pjpeg
Content-Length: 2712



----------7e13971310878
Content-Disposition: form-data;name="detectionPicture.jpg";filename="
detectionPicture.jpg";

Content-Type: image/pjpeg
Content-Length: 57899




You can get ANPR information like license plate number, vehicle type and vehicle color from XML

Value of vehicle type:
0-other vehicle, 1-small-sized vehicle, 2-large-sized vehicle, 3-pedestrian trigger, 4-two wheeler trigger, 5- tricycle

Valude of vehicle color:
"unknown", "white", "silver"-
silvery, "gray", "blacks"-balck, "red", "deepBlue"-dark blue, "blue", "yellow", "green", "brown", "pink", "purple", "deepGray"-dark gray, "cyan", "orange"

code of vehicle logo:

| Code | Main brand |
|---|---|
| 1026 | ALFAROMEO |
| 1027 | ASTONMARTIN |
| 1028 | AUDI |
| 1030 | PORSCHE |
| 1031 | BUICK |
| 1032 | BJQICHE |
| 1033 | BQZHIDAO |
| 1034 | BQWEIWANG |
| 1035 | BQYINXIANG |

| | |
|---|---|
| 1036 | BENZ |
| 1037 | BMW |
| 1038 | BAOJUN |
| 1039 | BAOLONG |
| 1040 | BENTLEY |
| 1041 | BRABUS |
| 1043 | HONDA |
| 1044 | PEUGEOT |
| 1045 | BYD |
| 1046 | CHANGHE |
| 1048 | GREATWALL |
| 1049 | CHANGAN |
| 1050 | DS |
| 1051 | SOUEAST |
| 1053 | VOLKSWAGEN |
| 1054 | DADI |
| 1056 | DODGE |
| 1059 | DAIHATSU |
| 1060 | TOYOTA |
| 1063 | FEREARI |
| 1064 | FORD |
| 1066 | FUDI |
| 1067 | FIAT |
| 1069 | MITSUOKA |
| 1070 | GZYUNBAO |
| 1071 | GQCHUANQI |
| 1074 | QOROS |
| 1076 | HUAPU |
| 1077 | HUATAI |
| 1078 | HAFEI |
| 1079 | HUMMER |
| 1080 | HAIMA |
| 1081 | HONGQI |
| 1083 | GEELYAUTO |
| 1084 | JEEP |
| 1085 | JAGUAR |
| 1086 | JIANGNAN |
| 1088 | CHRYSLER |
| 1089 | CADILLAC |
| 1091 | KANDIONE |
| 1093 | LAMBORGHINI |
| 1094 | LIFAN |
| 1095 | ROLLSROYCE |
| 1096 | LINCOLN |

| | |
|---|---|
| 1097 | EVERUS |
| 1098 | LIANHUA |
| 1100 | LOTUS |
| 1101 | LANDROVER |
| 1102 | SUZUKI |
| 1103 | LUFENG |
| 1104 | LEXUS |
| 1105 | RENAULT |
| 1107 | MINI |
| 1108 | MASERATI |
| 1109 | MEIYA |
| 1110 | MCLAREN |
| 1111 | MAYBACH |
| 1112 | MAZDA |
| 1114 | LUXGEN |
| 1115 | NJJINLONG |
| 1116 | OPEL |
| 1117 | ACURA |
| 1119 | VENUCIA |
| 1120 | CHERY |
| 1121 | KIA |
| 1123 | NISSAN |
| 1124 | RUIQI |
| 1125 | ROEWE |
| 1127 | SMART |
| 1128 | MITSUBISHI |
| 1129 | SQDATONG |
| 1131 | SHUANGHUAN |
| 1132 | SHUANGLONG |
| 1133 | SUBARU |
| 1134 | SKODA |
| 1135 | SAAB |
| 1138 | TIANMA |
| 1139 | TEALA |
| 1141 | DENZA |
| 1143 | WEILIN |
| 1144 | VOLVO |
| 1145 | WCYINGZHI |
| 1146 | XINKAI |
| 1147 | XINDADI |
| 1148 | XINYATU |
| 1149 | HYUNDAI |
| 1150 | SEAT |
| 1151 | CHEVROLET |

| | |
|---|---|
| 1152 | CITROEN |
| 1154 | YONGYUAN |
| 1156 | INFINITI |
| 1157 | MUSTANG |
| 1159 | YUJIE |
| 1160 | ZXAUTO |
| 1161 | ZHONGHUA |
| 1163 | ZOTYE |
| 1164 | KNOWBEANS |
| 1165 | KAIYI |
| 1166 | HUASONG |
| 1167 | JXWUSHILING |
| 1168 | BORGWARD |
| 1169 | SQTONGJIA |
| 1170 | HANJIANG |
| 1171 | ZINORO |
| 1172 | LUDIFANGZHOU |
| 1173 | HANTENG |
| 1175 | CHANGJIANG |
| 1176 | SWM |
| 1177 | KEYTON |
| 1180 | BISU |
| 1181 | CAKUAYUE |
| 1537 | ANKAI |
| 1538 | ANYUAN |
| 1540 | BBZHONGQI |
| 1546 | CHENGGONG |
| 1547 | CHANGLONG |
| 1549 | CASHANGYONG |
| 1552 | DONGFENG |
| 1554 | DAEWOO |
| 1555 | DAYUN |
| 1556 | DIMA |
| 1557 | DONGWO |
| 1559 | FUTIAN |
| 1561 | GMC |
| 1562 | GQJIAO |
| 1566 | HUALING |
| 1570 | HUIZHONG |
| 1571 | HIGER |
| 1574 | HTYUANTONG |
| 1575 | HANGTIAN |
| 1576 | HUANGHAI |
| 1577 | HEIBAO |

| | |
|---|---|
| 1578 | JIULONG |
| 1579 | JIANGHUAI |
| 1580 | JIANGHUAN |
| 1581 | JIANGLING |
| 1584 | JINBEI |
| 1585 | JINLONG |
| 1586 | KAIMA |
| 1587 | KAWEI |
| 1588 | KAIRUI |
| 1590 | LIANHE |
| 1592 | MAN |
| 1594 | NONGYONGCHE |
| 1596 | NANJUN |
| 1597 | QINGLING |
| 1598 | YOUNGMANONE |
| 1599 | SYZHONGGONG |
| 1600 | SHSHITONG |
| 1602 | TRICYCLE |
| 1603 | SQYWKHY |
| 1606 | SHAOLIN |
| 1608 | SHIFENG |
| 1609 | SUNWIN |
| 1611 | SHENYE |
| 1612 | SHUCHI |
| 1613 | SHANQI |
| 1614 | SCANIA |
| 1615 | TANGJUN |
| 1619 | WANFENG |
| 1620 | WUZHENG |
| 1621 | WULING |
| 1626 | XUGONG |
| 1629 | FAW |
| 1630 | YAXING |
| 1631 | IVECO |
| 1633 | YUTONG |
| 1634 | YANGZI |
| 1635 | YANTAI |
| 1636 | YUEJIN |
| 1637 | YINTIAN |
| 1639 | ZGZHONGQI |
| 1641 | ZHONGTONGONE |
| 1642 | ZHONGSHUN |
| 1644 | ZHONGDA |
| 1646 | JGZHONGKA |

| | |
|---|---|
| 1647 | WUZHOULONG |
| 1648 | COACH |
| 1651 | PICKUP |
| 1654 | JIJIANG |
| 1674 | DONGFANGHONG |
| 1676 | QINGQI |
| 1677 | TRUCK |
| 1678 | SPYCAR |
| 1679 | TRAILCAR |
| 1683 | GUILIN |
| 1684 | SCHYUNDAI |
| 1688 | WANXIANG |
| 1690 | LFSHIJUN |
| 1691 | CHANGAN |
| 1692 | ZLZHONGGONG |
| 1693 | YINLONG |
| 1695 | YIXING |
| 1696 | XIWO |
| 1697 | YANGZIJIANG |
| 1698 | SUITONG |
| 1702 | ZHONGTIANFC |
| 1703 | WANDA |
| 1704 | SHANGRAO |
| 1705 | ZHONGZHI |
| 1706 | ZCSDDIANDONG |
| 1707 | ZHONGTONGTWO |
| 1708 | GLCOACH |
| 1709 | BEIJING |
| 1710 | BEIFANG |
| 1711 | BFNAPULAN |
| 1712 | HUACHUAN |
| 1713 | YOUYI |
| 1714 | TONGXIN |
| 1715 | MG |
| 1716 | JIACHUAN |
| 1717 | NVSHEN |
| 1718 | SHILI |
| 1719 | SHAOLINTWO |
| 1720 | CHUANJIAO |
| 1721 | CHUANMA |
| 1722 | GUANGQI |
| 1723 | GQRIYE |
| 1724 | KANGDITWO |
| 1725 | HENGTIAN |

1726 HENGTONG
1727 XINFUDA
1728 XINLONGMA
1729 CHUANLAN
1730 CHUFENG
1731 JMCJM
1732 JMCZQ
1733 HAIOU
1734 MUDAN
1735 LEOPAARD
1736 SHENLONG
1737 FOTONSD
1738 HONGXING
1739 SHUCHITWO
1740 SHUDU
1741 HENGSHAN
1742 YUEXI
1743 YUANCHENG
1744 JINLV
1745 CAOUSHANG
1746 YOUNGMANTWO
1747 LINK
1748 FEIDIE
1749 FEICHI
1750 LISHAN
1751 JUNWEI
1752 NANQI
1753 DAHAN
1754 CHUNZHOU
1755 DIANKA
1756 HTWANSHAN
1757 ZOBENZFC
1758 YUNDU
1759 JUNMA
1760 GUOJIN
1761 WEIMA
1762 OULA
1763 NIO
1764 LADA
1765 JETOUR
1766 FORO
1767 HICOM
1768 JAC
1769 JEEP

| | |
|---|---|
| 1770 | JEEP |
| 1771 | PERODUA |
| 1772 | UD |
| 1773 | TOYOTA |
| 1774 | TOYOTA |
| 1775 | ISUZU |
| 1776 | HYUNDAI-ROHENS |
| 1777 | BEIBEN TRUCK |
| 1778 | SSANGYONG |
| | SSANGYONG-OLD |
| 1779 | VERSION |
| 1780 | HAVAL |
| 1781 | DAIHATSU |
| 1782 | DAEWOO |
| 1783 | PROTON |
| 1784 | PROTON |
| 1785 | PROTON |
| 1786 | GEELY-EMGRAND |
| 1787 | HINO |
| 1788 | UNKNOW |
| 1789 | KIA |
| 1790 | KIA-OLD VERSION |
| 1791 | KIA-BORREGO |
| 1792 | ALFA |
| 1793 | EQUUS-OLD VERSION |
| 1794 | RENAULT SAMSUNG |
| 1795 | MALAYSIA_UNKNOWN 1 |
| 1796 | AUCHAN |
| 1797 | BONLUCK |
| 1798 | JDMC |
| 1799 | WANXIANG |
| 1800 | SATE |
| 1801 | FLM |
| 1802 | SRM |
| 1803 | GEOMETRY |
| 1804 | BAOJUN-RS-5 |